

CITY UNIVERSITY OF HONG KONG

Course code & title : IS5411 Systems Analysis and Design

Session : Semester A 2020/21

Time allowed : Two hours

This paper has FOUR pages (including this cover page).

1. This paper consists of a case (Section A) and 4 questions (Section B).
 2. Questions in Section B ask you to develop UML models for the case in Section A. Answer ALL questions in Section B. Please draw the diagrams on the answer book.
 3. This is a closed-book exam. You are NOT allowed to bring your textbook, class notes, or any other reference materials.
 4. This exam is an individual assignment. You need to complete the exam independently. You are NOT allowed to collaborate with anybody on this exam.
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*This is a **closed-book** examination.*

No materials or aids are allowed during the whole examination. If any unauthorized materials or aids are found on a candidate during the examination, the candidate will be subject to disciplinary action.

Section A

BMuniverse.com is a website that buys used books and magazines and sells them to people who need them. Suppose you are hired by BMuniverse.com to develop a new website for the company. The requirements of the new website are described in the following narrative.

For a person to offer books for sale, he or she must register with BMuniverse.com. The person must provide a current physical address, telephone number, and an e-mail address. The system will then maintain an open account for this person. Access to the system as a seller will be through a secure, authenticated portal.

A seller can list books on the website through a special Internet form. Information required includes all of the pertinent information about the book including its category, the general condition, and the asking price. A seller may list as many books as desired. The website will maintain an index of all books so that buyers can use the search engine to search for books. The search engine will allow searches by title, author, category, and keyword.

People who want to buy books can search for the books they want. When they decide to buy, they must open an account with a credit card to pay for the books. The website will maintain all of this information on secure servers.

When a request to purchase is made, along with the payment, BMuniverse.com will send an e-mail notice to the seller of the book that is chosen. It will also mark the book as sold. The system will maintain this as an open order until it receives notice that the books have been shipped. After the seller receives notice that a listed book has been sold, he or she must notify the buyer via e-mail within 48 hours that the purchase is noted. Shipment of the order must be made within 24 hours after the seller sends the notification e-mail. The seller will send a notification to both the buyer and BMuniverse.com when the shipment is made.

After receiving notice of shipment, BMuniverse.com will maintain the order in a shipped status. At the end of each month, a check will be mailed to each seller for the book orders that have been in a shipped status for 30 days. The 30-day waiting period is to allow the buyer to notify BMuniverse.com if the shipment does not arrive for some reason, or if the book is not in the same condition as advertised.

The buyers can, if they want, submit feedback on the seller's service. Some sellers might be very active and may use BMuniverse.com as a major outlet for selling books. So, buyers' feedback is important information to potential buyers.

Section B

Question 1 (20 points): Figure 1 shows you a use case diagram for BMuniverse.com. Draw two activity diagrams for use cases in Figure 1: one for **Register** (a seller) and one for **Purchase a book**. In the two diagrams, you need to show (1) all the actors involved in these use cases (**5 points**) and (2) the flows of activities (**10 points**). Your diagram must follow the UML conventions (**5 points**).

Note: **Register** (a seller) is shown on the top of Figure 1. It is important to clearly show the flow of activities. Missing important details in the diagrams will result in significant loss of points.

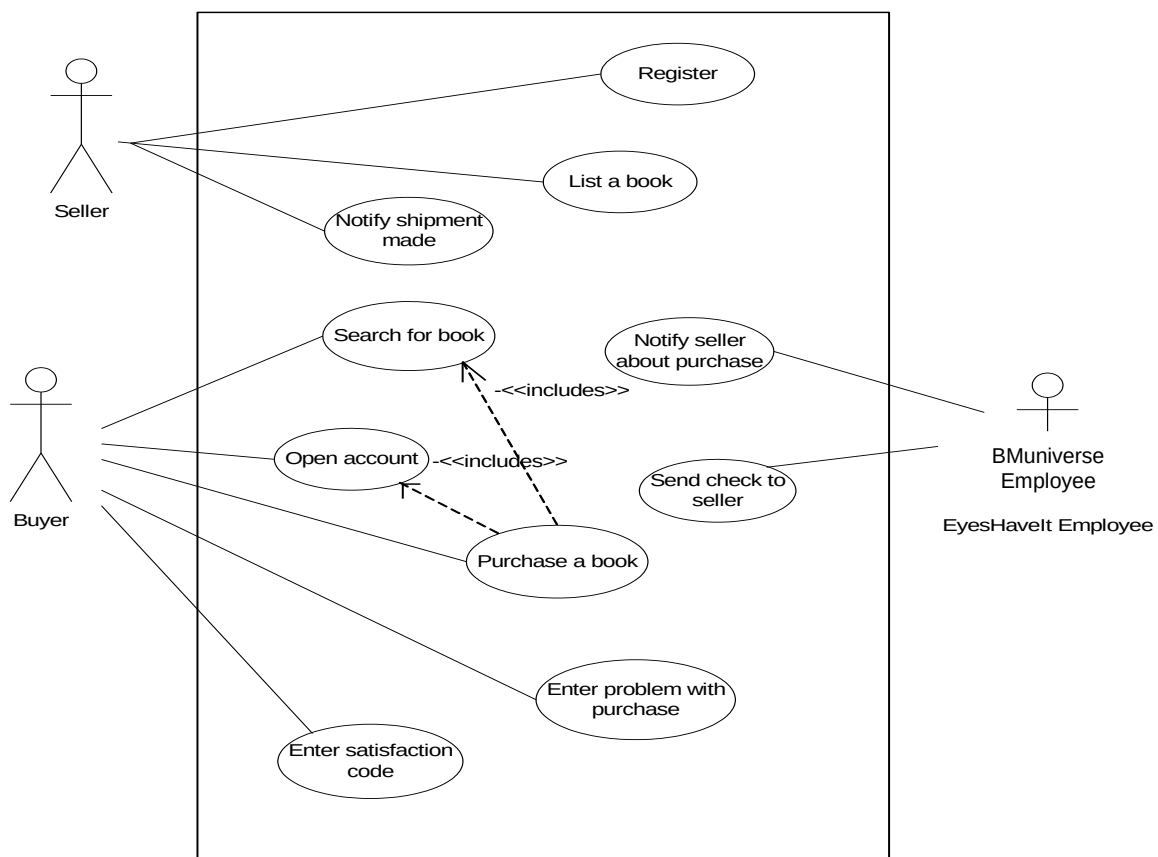


Figure 1 Use Case Diagram

Question 2 (25 points): Develop an analysis class diagram for BMuniverse.com. In this diagram you need to show (1) all the data entities involved in this business domain (**5 points**), (2) necessary attributes for each data entity (**5 points**), and (3) associations between the data entities with multiplicity (i.e. one-to-one, one-to-many relationships etc.) (**10 points**). Your diagram must follow the UML conventions (**5 points**).

Question 3 (30 points): Develop a first-cut design class diagram for the use case *Register* (a seller). In this diagram you need to show (1) all the classes involved in this use case (*5 points*), (2) all the attributes, their visibility, and their data type for each class (*5 points*), (3) the navigation visibilities (i.e. reference relationships) between these classes (*10 points*), and (4) the controller class (*5 points*). (5) Your diagram must also follow the UML conventions (*5 points*).

Question 4 (25 points): Develop a design system sequence diagram (SSD) for the use case *Register* (a seller). In this diagram you need to show (1) the interactions between seller and the system (*5 points*), (2) all the necessary objects (including view layer objects and data access objects) inside the system to handle seller's request (*5 points*), and (3) all the interactions between the objects within the system to handle seller's request (*10 points*). (4) Your diagram must follow the UML conventions (*5 points*). You need to cover as many details as you could in the design. A diagram that does not clearly show the necessary details will result in significant loss of points.